Curriculum Vitae of Lambodhar Damodaran

Department of Biology
Emory University

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Education

2018 – 2023 University of Georgia - Institute of Bioinformatics
Ph.D. Bioinformatics
IDEAS (Interdisciplinary Disease Ecology Across Scales) NSF-NRT Graduate Research Fellow

Thesis title: Molecular Epidemiology of Contemporary Seasonal influenza Epidemics

Major Professor: Justin Bahl Ph.D.
Committee members: Andrew Park Ph.D.
Liliana Salvador Ph.D.
Stephen Mark Tompkins Ph.D.

2014 – 2018 University of North Carolina at Charlotte
Major: B.S. Biology
Minors: Bioinformatics and Genomics, Biotechnology

Research Experience

2025 – Present Postdoctoral Fellow - Katia Koelle Lab - Emory University, Department of Biology

2023 – 2025 Postdoctoral Fellow – Louise Moncla Lab – University of Pennsylvania School of Veterinary Medicine, Department of Pathobiology

2018 – 2023 Graduate Research Assistant - Justin Bahl Lab - University of Georgia, Institute of Bioinformatics

Summer 2021 Research Internship – Influenza Division – Centers for Disease Control

Performed research with the Virology, Surveillance, and Diagnostics Branch in the Influenza Division at the Centers for Disease Control under the guidance of Dr. Rebecca Kondor.

2015 – 2018 Undergraduate Researcher - Janies Lab – UNC Charlotte

Teaching Experience

Classes

Spring 2026	UNC Charlotte – BIOL 4242 – Biology of Birds lecture
Fall 2025	Emory University - BIOL 463 – Molecular Epidemiology lecture
2016 - 2018	Teaching assistant – Department of Biology - UNC Charlotte General Biology I & II

Seminars

2024 – 2025	Penn First Exposure to Research in the Biological Sciences (FERBS) University of Pennsylvania Mentored students of the FERBS program at UPenn which places freshman students from underrepresented backgrounds in research labs and gives them opportunities to learn about the scientific process.
September 2019	International Bioinformatics Workshop on Molecular Epidemiology and Phylodynamics Organizers: Johns Hopkins CEIRS, Linkou Chang Gung Memorial Hospital, Chang Gung University – Taoyuan, Taiwan – September 16-20th 2019. Taught public health practitioners Bayesian phylodynamic workflows for different genomic datasets.

Honors/Awards

2023 - 2024	Fellow-In-Residence – Verena Institute NSF Biology Integration Institute - Georgetown University Collaborating PI: Dr. Barbara Han (Cary Institute of Ecosystem Studies)
2019 - 2022	IDEAS (Interdisciplinary Disease Ecology Across Scales) NSF-NRT Graduate Research Fellowship Odum School of Ecology - University of Georgia
2014 - 2018	University Honors Program – UNC Charlotte
2017	Charlotte Research Scholar – UNC Charlotte

Skills/ Proficiencies

Programming Languages and Computational skills:

Java, R, Python, Bash, Unix proficiency, HPC cluster systems proficiency (SLURM, PBS, LSF), Git, SQL, PostgreSQL. Github: ldamodaran

Genomics skills:

Genome assembly (IRMA,SPAdes, nf-core viralrecon), metagenomics (nf-core mag), databasing proficiency, GWAS, transcriptomics, taxonomic classification (Kraken2)

Phylogenetics skills:

Maximum likelihood phylogenetic reconstruction (RAxML, IQTREE, etc.), Bayesian phylogenetic reconstruction (BEAST 1 and 2)

Wet lab skills:

PCR, RT-qPCR, DNA extraction (FTA Blood card, arthropods), Sanger sequencing.

Languages:

Native proficiency: English, Tamil, Telugu
Limited proficiency: Spanish, French

Journal Reviewer – *ad hoc*:

Proceedings of the Royal Society B, PLOS Pathogens. Virus Evolution, BMC Microbiology, International Journal of Infectious Disease, NPJ Viruses, Scientific Reports, Infectious Diseases of Poverty, One Health Outlook, Tropical Medicine and Health

Invited talks

May 2026, 33rd International Dynamics and Evolution of Human viruses, Kelowna BC

October 2025: Population Biology, Ecology and Evolution lecture series, Emory University

March 2025: Larry Mays Seminar in Bioinformatics, University of North Carolina at Charlotte

March 2025: Phylodynamics for Canadian Public Health, Public Health Department of Ontario/ University of Toronto

December 2024: Computational modeling core meeting, Centers for Excellence in Influenza Research and Response (Virtual)

Publications

Lambodhar Damodaran, Amber Coats, Jack Lin, Michael Martin, and Katia Koelle. "Evaluating the role of prolonged infections in the emergence of norovirus GII.4 variants" (In prep)

Rivero, Ricardo, David Simons, **Lambodhar Damodaran**, Irene Karegi, Sarah Gurev, Daniel J. Becker, Dan L. Warren, Nicola F. Mueller, David A. Rasmussen, and Stephanie N. Seifert. "Molecular and ecological determinants of effective reassortment in orthohantaviruses." *bioRxiv* (2026): 2026-06 (In Review, Nature Ecology and Evolution)

Maltepes, Maria A., Alexey Markin, Tavis K. Anderson, Kathryn Kistler, Grant Park, **Lambodhar Damodaran**, Jordan Ort, Jared Sabre, Stephen Shank, and Louise H. Moncla. "Host type governs influenza evolutionary strategy across reservoir and spillover hosts." *bioRxiv* (2026): 2026-09. (In Review, Nature Microbiology)

Lambodhar Damodaran, Joseph A. Lewnard, Gregg S. Davis, Sara Y. Tartof, Louise H. Moncla, and Nicola F. Müller. "Frequent seasonal reassortment between high and low path viruses drives the diversification of influenza A/H5N1." *bioRxiv*(2026): 2026-04. (In Review, PLOS Pathogens)

Hugo G. Castelan, **Lambodhar Damodaran**, Art Poon. "Bayesian phylodynamic analysis reveals successful lineage expansion and surveillance gaps in Human Parechoviruses" (In Review, Journal of General Virology)

Max SY Lau, Leke Lyu, Nicola Mueller, Zewen Liu, **Lambodhar Damodaran**, and Wei Jin. "Rapid Phylogeographic Inference of Regional Epidemic Dynamics for Routine Genomic Surveillance." (In Revision, PNAS).

Lambodhar Damodaran, Anna Jaeger, and Louise H. Moncla. "Ecology and spread of the North American H5N1 epizootic." *Nature* 649, 432–441 (2026).

Lambodhar Damodaran, Cody Dailey, Swan Tan, Norman Hassell, Rebecca Kondor, Ludy R. Carmola, John M. Drake, Pejman Rohani, Justin Bahl. "Analysis of seasonal H3N2 influenza diffusion in the United States, 2011-2020." (In revision, *Influenza and Other Respiratory Viruses*)

Alpha Forna, **Lambodhar Damodaran**, Christian E. Gunning, Parnian Rahimi, Aarya Venkat, Natarajan Kannan, Rebecca Kondor, Justin Bahl, Pejman Rohani, and John M. Drake. "Alignment-free prediction of cross-reactivity in influenza A (H3N2) anticipates antigenic drift." *PLOS Computational Biology* 22, no. 8 (2026): e1014628.

Ricardo Rivero, Vaneza Tique, **Lambodhar Damodaran**, Daniel Echeverri De la Hoz, Daniela Paternina, Mauricio Santos-Vega, Verity Hill, Nathan D. Grubaugh, and Salim Mattar." Genomic epidemiology of dengue virus 2 and 3 reveals repeated introductions and exportations of several lineages in Colombia." *Cell Reports* 45, no. 1 (2026).

Lora Lee Pless, **Lambodar Damodaran**, Ray Pomponio, Rose Patrick, Marissa Pacey Griffith, Sara Walters, Kady D Waggle, Atalia Pleskovitch, Vatsala Rangachar Srinivasa, Cole A Varela, Lee H Harrison, John P Barton, Louise H Moncla, Marian G Michaels, John V Williams, Anna F Wang-Erickson "Emergence and epidemiology of dominant variants of human metapneumovirus in the United States between 2016 and 2021". *mBio* 17:e02619-25 (2026).

Lyu, Leke, Gabriella Veytsel, Guppy Stott, Spencer Fox, Cody Dailey, **Lambodhar Damodaran**, Kayo Fujimoto, Pamela Brown, Roger Sealy, Armand Brown, Magdy Alabady, Justin Bahl. "Characterizing spatial epidemiology in a heterogeneous transmission landscape using the spatial transmission count statistic." *Communications Medicine* 5, no. 1 (2025): 165.

Brian R. Wasik, **Lambodhar Damodaran**, Maria A. Maltepes, Ian EH Voorhees, Christian M. Leutenegger, Sandra Newbury, Louise H. Moncla, Benjamin D. Dalziel, Laura B. Goodman, and Colin R. Parrish. "The evolution and epidemiology of H3N2 canine influenza virus after 20 years in dogs." *Epidemiology & Infection* 153 (2025): e47.

Alpha Forna, Bodie Weedop, **Lambodhar Damodaran**, Norman Hassell, Rebecca Kondor, Justin Bahl, John M. Drake, and Pejman Rohani. "Sequence-based detection of emerging antigenically novel influenza A viruses." *Proceedings B* 291, no. 2028 (2024): 20240790.

Martin Grunnill, Alireza Eshaghi, **Lambodhar Damodaran**, Sandeep Nagra, Ali Gharouni, Thomas Braukmann, Shawn Clark, Adriana Peci, Sandra Isabel, Philip Banh, Louis du Plessis, Carmen Lia Murall, Caroline Colijn, Samira Mubareka, Maan Hasso, Justin Bahl, Heba H. Mostafa, Jonathan B. Gubbay, Samir N. Patel, Jianhong Wu, Venkata R. Duvvuri. "Inferring enterovirus D68 transmission dynamics from the genomic data of two 2022 North American outbreaks." *npj Viruses* 2, no. 1 (2024): 34.

Ian Padykula, **Lambodhar Damodaran**, Kelsey T. Young, Madelyn Krunkosky, Emily F. Griffin, James F. North, Peter J. Neasham, Vasilis C. Pliasas, Chris L. Siepker, James B. Stanton, Elizabeth W. Howerth, Justin Bahl, Constantinos S. Kyriakis, and Stephen Mark Tompkins. "Pandemic Risk Assessment for Swine Influenza A Virus in Comparative In Vitro and In Vivo Models." *Viruses* 16, no. 4 (2024): 548.

Jonathon D. Gass Jr,, Nichola J. Hill, **Lambodhar Damodaran**, Elena N. Naumova, Felicia B. Nutter, Jonathan A. Runstadler. Ecogeographic drivers of the spatial spread of highly pathogenic avian influenza outbreaks in Europe and the United States, 2016–Early 2022. *International Journal of Environmental Research and Public Health* 20, no. 11 (2023): 6030.

Venkata R. Duvvuri, Joseph T. Hicks, **Lambodhar Damodaran**, Martin Grunnill, Thomas Braukmann, Jianhong Wu, Jonathan B. Gubbay, Samir N. Patel, Justin Bahl. Comparing the transmission potential from sequence and surveillance data of 2009 North American influenza pandemic waves. *Infectious Disease Modelling* (2023) https://doi.org/10.1016/j.idm.2023.02.003

Tamar E Carter, Araya Gebresilassie, Shantoy Hansel, **Lambodhar Damodaran**, Callum Montgomery, Victoria Bonnell, Karen Lopez, Daniel Janies, Solomon Yared. Comparison of the

knockdown resistance locus (kdr) in Anopheles stephensi, An. arabiensis, and Culex pipiens s.l. suggests differing mechanisms of pyrethroid resistance in east Ethiopia. *Journal of the American Society for Tropical Medicine and Hygiene* (2022) doi: 10.4269/ajtmh.20-1357

Kayo Fujimoto, Justin Bahl, Joel O. Wertheim, Natascha Del Vecchio, Joseph Hicks, **Lambodhar Damodaran**, Camden J. Hallmark, Richa Lavingia, Ricardo Mora, Michelle Carr, Biru Yang, John A. Schneider, Lu-Yu Hwang, Marlene McNeese. Methodological synthesis of Bayesian phylodynamics, HIV-TRACE, and GEE: HIV-1 transmission epidemiology in a racially/ethnically diverse Southern U.S. context. *Scientific Reports* (2021)

Solomon Yared, Araya Gebressielasie, **Lambodhar Damodaran**, Victoria Bonnell, Karen Lopez, Daniel Janies, and Tamar E. Carter. Insecticide resistance in Anopheles stephensi in Somali Region, eastern Ethiopia. *Malaria journal* 19, no. 1 (2020): 1-7.

Lambodhar Damodaran, Adriano de Bernardi Schneider, Shi Chen, Daniel Janies. Evolution of endemic and sylvatic lineages of dengue viruses. *Cladistics* (2019)

Tamar E. Carter, Solomon Yared, Araya Gebresilassie, Victoria Bonnell, **Lambodhar Damodaran**, Karen Lopez, Mohammed Ibrahim, Seid Mohammed, and Daniel Janies. "First detection of Anopheles stephensi Liston, 1901 (Diptera: culicidae) in Ethiopia using molecular and morphological approaches." *Acta tropica* 188 (2018): 180-186.

Tamar Carter, Daniel Janies, Karen Lopez, Victoria Bonnell, **Lambodhar Damodaran**, Abraham Aseffa, Seleshi Kebede Mekonnen. (2017) Glucose-6-phosphate dehydrogenase deficiency (G6PD) genetic variants in malaria patients in southwestern Ethiopia. *Journal of the American Society for Tropical Medicine and Hygiene* (2017) doi: 10.4269/ajtmh.17-0557

Adriano Schneider, Robert W.M, Jun-Tao Guo, Jane Homan, Gregorio Linchangco, Zach Witter, Dylan Vinesett, **Lambodhar Damodaran**, Daniel Janies. Molecular evolution of Zika virus as it crossed the Pacific to the Americas. *Cladistics*. (2017) doi: 10.1111/cla.12178

Robert W.M, Jane Homan, Michael V Callahan, Jill Glasspool-Malone, **Lambodhar Damodaran**, Adriano Schneider, Rebecca Zimler, James Talton, Ronald R Cobb, Ivan Ruzic, Julie Smith-Gagen, James Wilson, Daniel Janies. Zika Virus: Medical Countermeasure Development Challenges. *PLoS Negl Trop Dis* 10(3): e0004530. (2016) doi: 10.1371/journal.pntd.0004530

Selected Abstracts presented:

Lambodhar Damodaran, Rustom Antia, and Katia Koelle. “Evaluation of a model of antigenic evolution in influenza A virus H3N2” 33rd International Dynamics and Evolution of Human Viruses, Kelowna BC, May 2026.

Angie Chen, **Lambodhar Damodaran**, and Katia Koelle. “Codon bias in seasonal influenza viruses reveal selection pressures and timescales of codon volatility acquisition”. 33rd International Dynamics & Evolution of Human Viruses, Kelowna BC (2026).

Elizabeth Somsen, **Lambodhar Damodaran**, and Katia Koelle. “Understanding patterns of drift and selection in the evolution of influenza hemagglutinin stability using phylogenetic comparative methods”. 33rd International Dynamics and Evolution of Human Viruses, Kelowna BC, May 2026.

Lambodhar Damodaran, Amber Coats, Jack Lin, Michael Martin, and Katia Koelle. **“**The role of prolonged infections in the emergence of norovirus GII.4 variants”. 33rd International Dynamics and Evolution of Human Viruses, Kelowna BC, May 2026. (**Talk**)

Anna F. Wang-Erickson, **Lambodar Damodaran**, Ray Pomponio, Rose Patrick, Marissa Pacey Griffith, Sara Walters, Kady D. Waggle, Atalia Pleskovitch, Vatsala Rangachar Srinivasa, Cole A. Varela, Lee H. Harrison, Louise H. Moncla, Marian G. Michaels, Lora Lee Pless, John V. Williams. “Emerging Dominant Variants of Human Metapneumovirus in the US”. Negative Strand Virus Meeting. Montpellier. France, June 2025.

Lambodhar Damodaran, Anna Jaeger, and Louise H. Moncla. "Intensive transmission in wild, migratory birds drove rapid geographic dissemination and repeated spillovers of H5N1 into agriculture in North America.". EEID Conference, Stanford. June 2024.

Maria Maltepes, **Lambodhar Damodaran**, Louise Moncla. Characterizing the adaptive landscape of H3Nx Influenza A viruses across diverse hosts. EEID Conference, Stanford. June 2024.

Ricardo Rivero, **Lambodhar Damodaran**, David Simons, Daniel J. Becker, Stephanie N. Seifert. Linking Genomic and Spatiotemporal Surveillance Datasets to Uncover the Evolutionary History of Hantaviruses. Idaho and Washington Chapter of The Wildlife Society 2024.

Lambodhar Damodaran, Cody Dailey, Swan Tan, Rebecca Kondor, Justin Bahl. Genetic and antigenic characterization of global seasonal influenza virus evolution, 2017-2022. 1st Annual CIERR meeting. August 2023.

Josanne H. Verhagen, Eline A.M. Vink, Marjolein J. Poen, Frank A. Majoor, Divya Kriti, Jayeeta Dutta, Harm van Bakel, Oanh Vuong, Rachel D. Scheuer, Mariëlle van Toor, Nicola S Lewis, Nichola N Hill, Justin Bahl, **Lambodhar Damodaran**, Jiani Chen, Thijs Kuiken, Ron A.M. Fouchier, Mart C.M. de Jong. An Epidemiology Model of Influenza A Virus in Wild Birds Based on Surveillance of Black-headed Gulls (*Chroicocephalus ridibundus*).12th International Congress For Veterinary Virology, September 2022.

Lambodhar Damodaran, Cody Dailey, Rebecca Kondor, Justin Bahl. Analysis of seasonal H3N2 influenza diffusion in the United States, 2011-2020. 14th Annual CEIRS meeting, August 2022.

Swan Tan, Jiani Chen, **Lambodhar Damodaran**, Cody Dailey, Rebecca Kondor, Lenny Moise, Anne S De Groot, Justin Bahl. Sequence-based evaluation of the immune landscape of seasonal influenza A(H3N2) virus. 14th Annual CEIRS meeting, August 2022.

Ian Padykula, **Lambodhar Damodaran**, Kelsey Young, Madelyn Krunkosky, Chris Siepker, James Stanton , Elizabeth Howerth , Justin Bahl, Constantinos S. Kyriakis, S. Mark Tompkins. Pandemic risk assessment for a swine influenza A virus in comparative human substrates. ASV 40th annual meeting. July 2021.

Lambodhar Damodaran, Kayo Fujimoto, Joseph Hicks, Natascha Brauchle, Camden Hallmark, Ricardo Mora, Michelle Carr, Marlene McNeese, Joel Wertheim, Justin Bahl. Inference of HIV-1 transmission dynamics through comparative genetic analysis of surveillance data. Epidemics 7 - International Conference on Infectious Disease Dynamics in Charleston, SC. December 2019.

Tamar Carter, **Lambodhar Damodaran**, Shantoy Hansel, Callum Montgomery, Karen Lopez, Daniel Janies, and Solomon Yared. Insecticide resistance gene mutations in Anopheles stephensi, An. arabiensis, and Cu. pipiens s.l. in eastern Ethiopia. *American Journal of Tropical Medicine and Hygiene*. Vol. 101. 8000 Westpark Dr, STE 130, McLean, VA 22101 USA: AMER SOC TROP MED & HYGIENE, 2019.

Adriano de Bernardi Schneider**, Lambodhar Damodaran**, Daniel Janies. (2017) "Transmission network analyses of infectious diseases outbreaks in the genomic era. Chemical and Biological Defense Science & Technology Conference November 2017 in Long Beach CA

Lambodhar Damodaran, Adriano de Bernardi Schneider, Daniel Janies. (2017) Phylogenetic Analysis of Sylvatic and Endemic Dengue virus Evolution. 66th American Tropical Medicine and Hygiene Annual meeting. November 5th- 11th 2017 in Baltimore MD

Tamar Carter, **Lambodhar Damodaran**, Daniel Janies. (2017) Plasmodium Falciparum Genetic Diversity and Malaria Transmission: A Meta-analysis Approach Using Microsatellite Data. Pathogen Genomics. Annual Meeting of the Society for Molecular Biology and Evolution 2017 (SMBE). July 2nd – 6TH 2017 in Austin TX

Daniel Janies, Adriano Schneider, **Lambodhar Damodaran,** Dylan Vinesett, Zachary Witter. (2016) Phylogeography and molecular evolution of Zika virus. International Zika Summit . April 2016, Institut Pasteur Paris, France.

Daniel Janies, Colby Ford, **Lambodhar Damodaran**, Zach Witter. (2016) Spread of Middle East Respiratory Coronavirus: Genetic versus Epidemiological Data. International Society for Disease Surveillance Conference December 6-9th 2016 in Atlanta, GA
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5462161/